



Assignment

Spring 2026 2nd Semester

Course Title: Circuit Analysis Lab

Course Code: CMPE-121L

Topic(s): Lab 11 Verification of Norton's Theorem

SUBMITTED TO:

Mr. Mehmood-ul-Haq

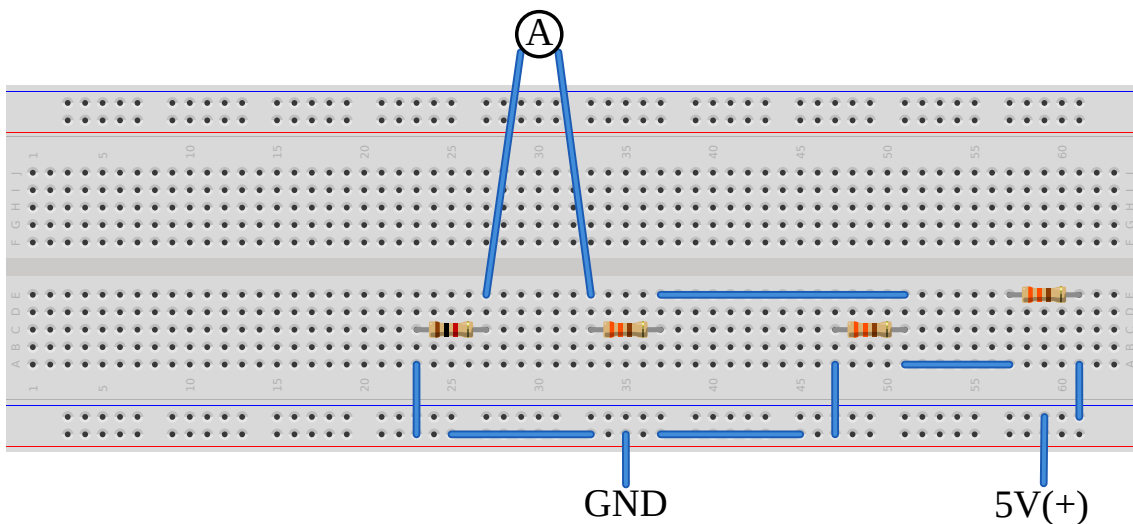
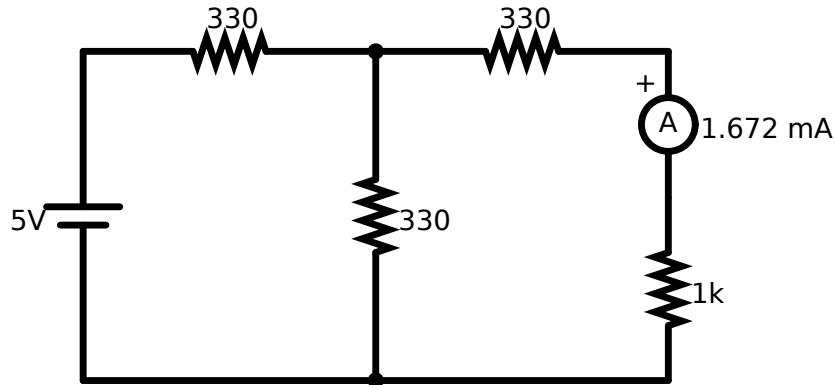
SUBMITTED BY:

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Submission Date: 19 May, 2026

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Experiment 1



Breadboard circuit layout: Initial assembly with load resistance $R_L = 1\text{ k}\Omega$

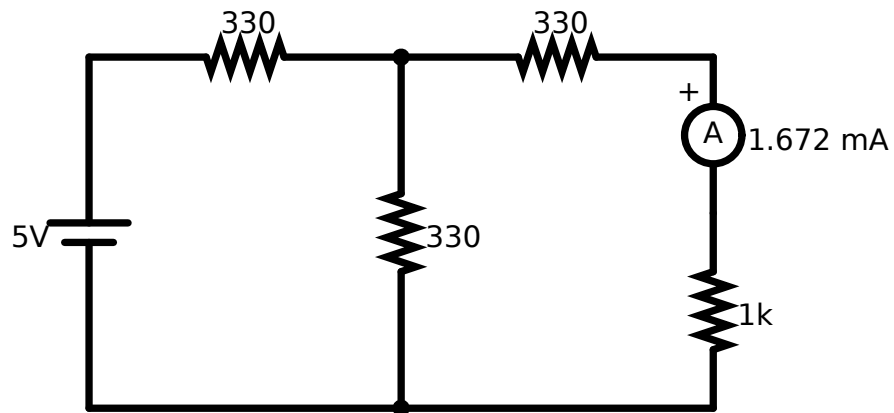
Practical Procedure

The objective of this experiment is to verify Norton's Theorem by simplifying a linear bilateral active two-terminal network into an equivalent circuit consisting of a single constant current source (I_N) in parallel with an equivalent resistance (R_N). The primary circuit is constructed using three 330Ω resistors and a $1\text{ k}\Omega$ load resistor (R_L), powered by a Regulated Power Supply. For all measurements, the **UNI-T UT89XE** digital multimeter is utilized to capture the load current,

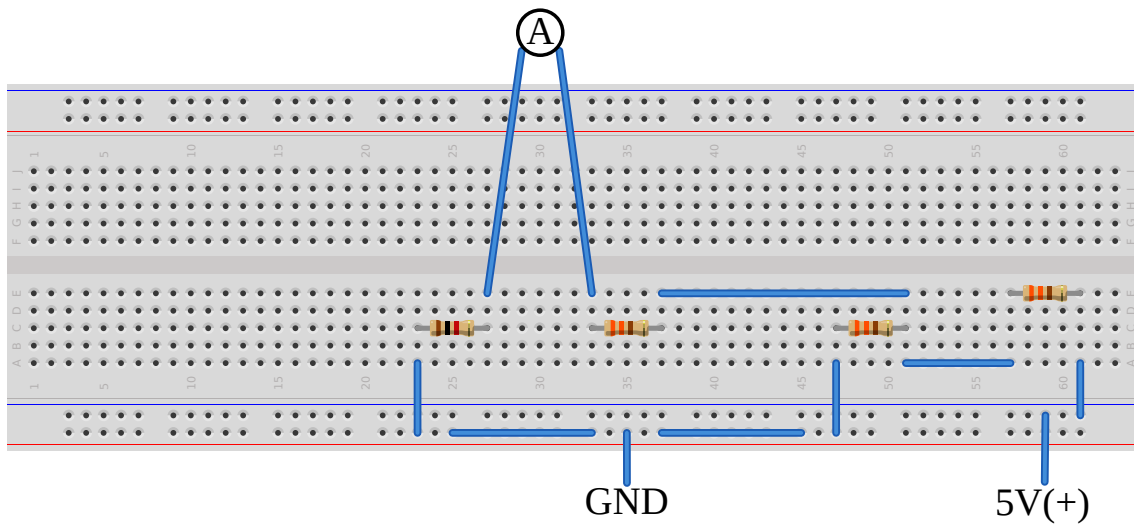
short-circuit current, and looking-back resistance. A supply voltage of 5 V is provided from the RPS to the initial circuit to observe the current while the load is connected.

To measure the load current (I_L), the multimeter dial is set to the **20mA DC** range (indicated by the **A**— symbol). The black probe is connected to the **COM** port while the red probe is inserted into the **mA μ A** terminal. This measurement requires the DMM to be connected in **series** with the load. To find the Norton current (I_N), the load resistor is removed and the terminals are short-circuited through the ammeter to measure the short-circuit current. To determine the Norton resistance (R_N), the RPS is removed and short-circuited; the DMM is then set to the **2000 Ω** range and probes are connected across the load terminals to measure the "looking-back" resistance.

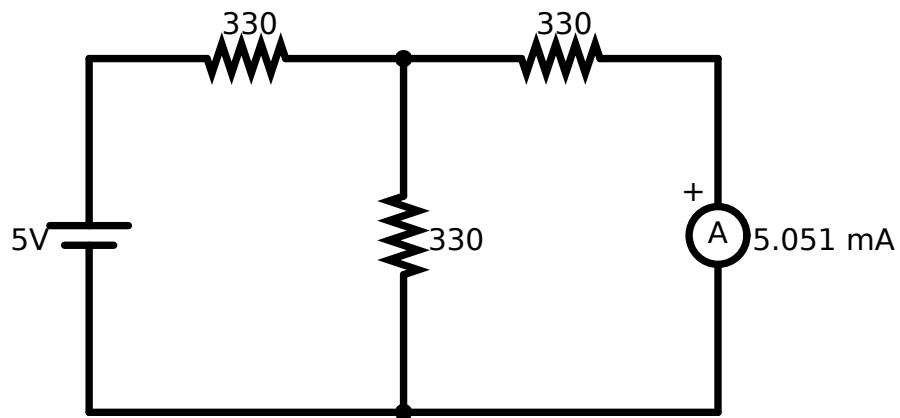
After obtaining I_N and R_N , the Norton equivalent circuit is constructed by setting the current source to the measured I_N value and placing a resistance approximating R_N in **parallel** with the 1 k Ω load resistor. The final objective is to verify whether the equivalent circuit produces the same load current (I_L) as recorded in the original network. Theoretical accuracy depends on component linearity; however, minor discrepancies may be observed in simulated or physical environments. Factors such as the **internal resistance** of the power supply and the **burden voltage** of the ammeter can introduce slight deviations between the mathematical predictions and the recorded experimental data.



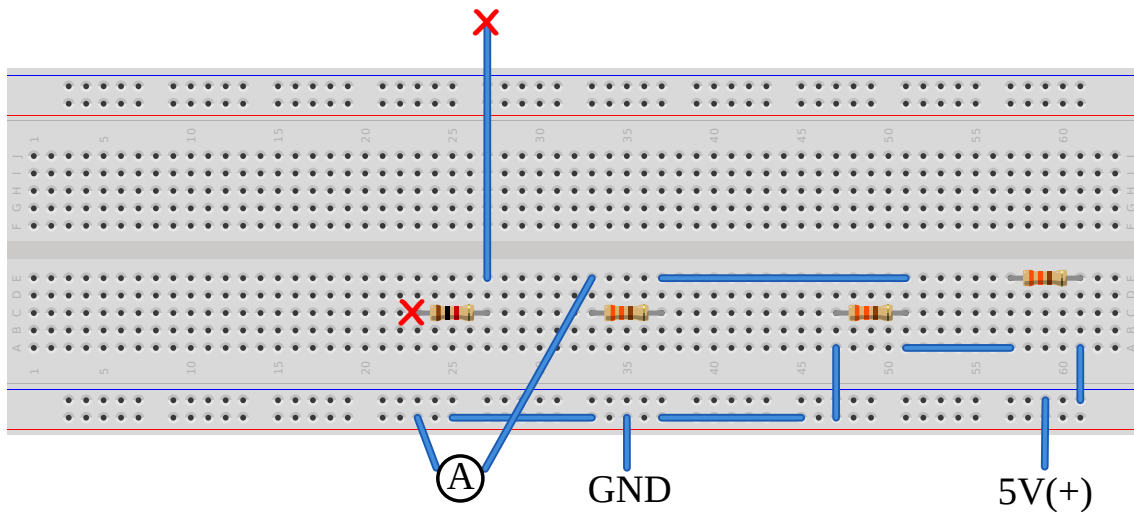
Falstad: Measurement of Load Current I_L in the original network



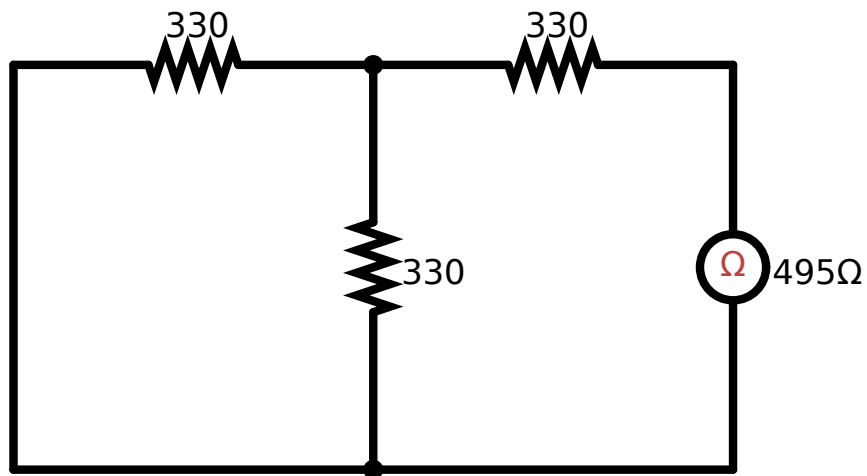
Breadboard: Series ammeter connection for I_L measurement



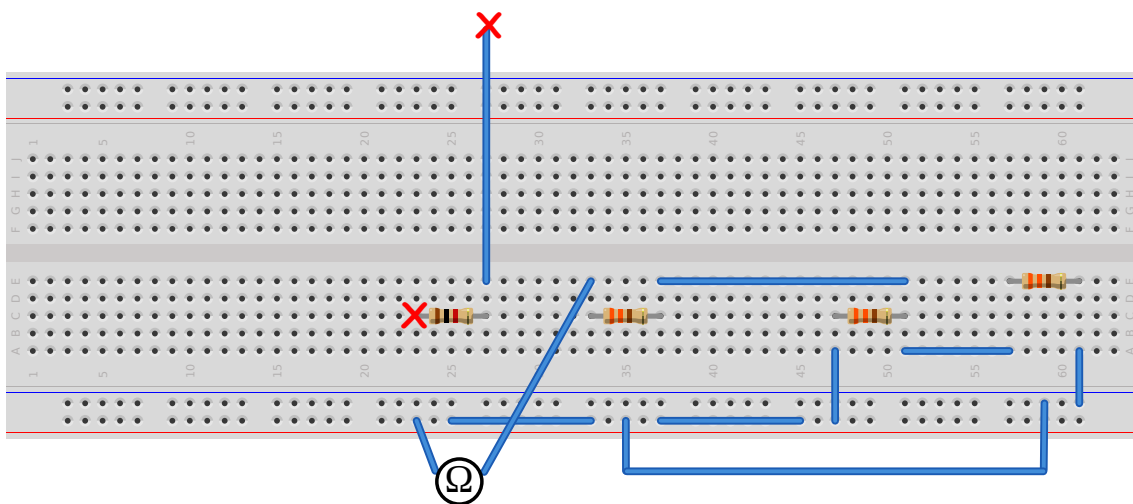
Falstad: Short-circuit current measurement for I_N



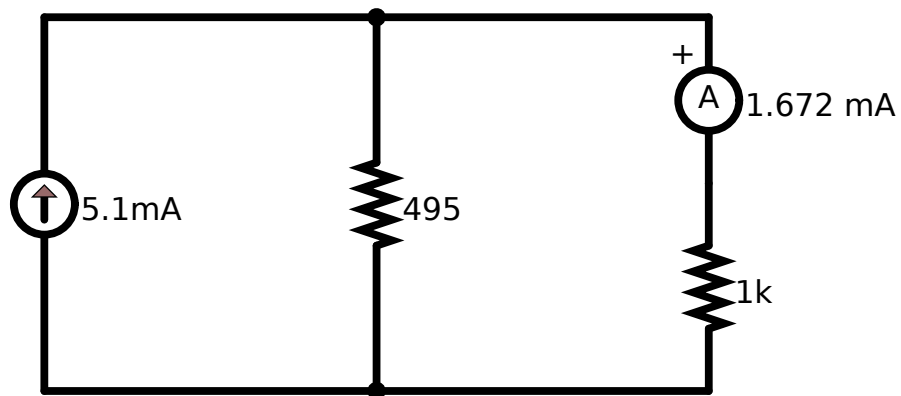
Breadboard: Ammeter shorting the load terminals to measure I_N



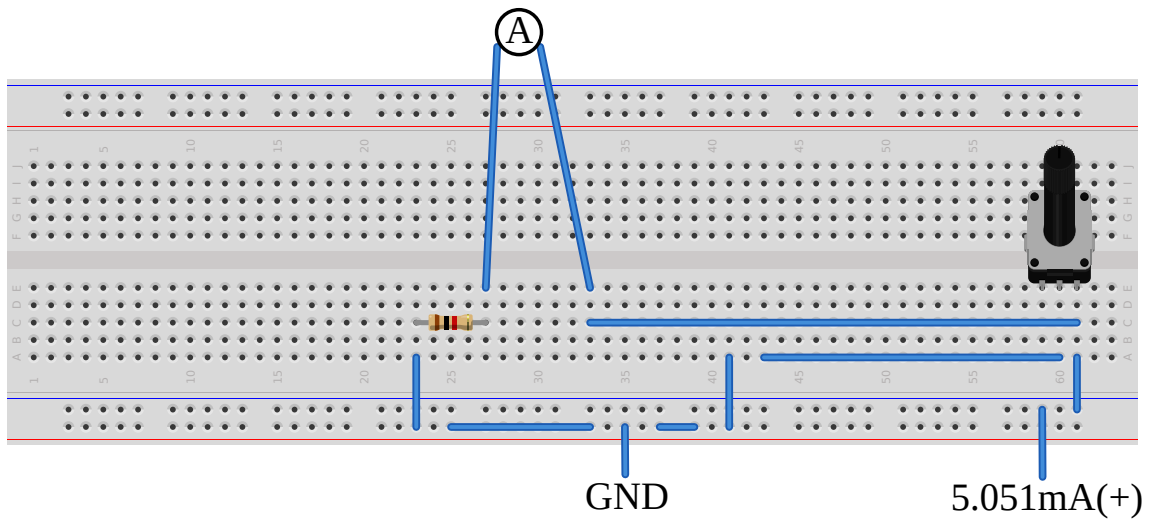
Falstad: Resistance measurement for R_N (Source shorted)



Breadboard: Ohmmeter connection for R_N calculation



Falstad: Verification of I_L using the Norton Equivalent Circuit



Breadboard: Equivalent circuit using a current source and parallel resistor combination

Experiment 1 Data

Practical Values

Sl. No.	Source E	I_N	R_N	Load Current I_L	
				Original Circuit	Equivalent Circuit
	Volts	mA	Ω	mA	mA
1	5.0	5.051	495	1.672	1.672

Post-Lab Questions

1. What is Norton's Theorem?

Norton's Theorem states that any linear, bilateral, active two-terminal network can be replaced by an equivalent circuit consisting of a single constant current source (I_N) in parallel with a resistance (R_N). The Norton current is the short-circuit current across the terminals, and the Norton resistance is the equivalent resistance seen from the terminals with all independent sources deactivated.

2. What are the applications of Norton's Theorem?

Norton's Theorem is primarily used for simplifying complex networks to analyze the behavior of varying loads. It is particularly useful in current-source modeling, transistor circuit analysis, and when performing source transformations between voltage and current equivalents to simplify nodal analysis.

3. What are the theoretical and practical limitations of Norton's Theorem?

Theoretically, the theorem only applies to linear and bilateral networks. Practically, ideal current sources are difficult to implement, and high-frequency effects can limit the accuracy of the resistive model. Additionally, measuring I_N requires a near-zero resistance ammeter to ensure a true short-circuit condition.